

# Northstar Labs — 01 Company

Fictional reference document created for RAG retrieval, chunking, embedding, vector search, metadata filtering, and multi-document QA testing.

## 1. Company Overview

Northstar Labs is a fictional technology company used only for RAG testing. It builds workflow automation, analytics, and AI knowledge software for mid-market organizations.

The company was founded in 2019. Its engineering groups are Platform, Applications, Data, AI Systems, and Reliability. Product teams own customer outcomes while platform teams provide shared infrastructure.

Testing note: This section contains source-of-truth fictional facts for semantic retrieval. Related concepts may appear in other documents with different details, allowing comparison of source relevance.

## 2. Architecture

Northstar separates synchronous APIs from asynchronous workflow execution.

PostgreSQL stores transactional entities such as users, organizations, workflows, and execution records. Redis supports caching, locks, and queues. Object storage holds large files and exports.

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### **3. Security**

Northstar applies organization-level data isolation and least-privilege access.

OAuth provider tokens are encrypted and must never appear in logs. Sensitive exports require authorization and audit records. Production secrets are supplied through managed secret storage.

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## 4. Reliability

The core API platform targets 99.9 percent monthly availability.

Latency is tracked at p50, p95, and p99. Production incidents use an incident commander, followed by a blameless review containing impact, timeline, contributing factors, and actions.

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## 5. AI Systems

Beacon is Northstar's fictional RAG knowledge assistant.

Documents are normalized, chunked, embedded, and indexed. Queries retrieve candidate passages using vector similarity and metadata filters. Retrieved evidence is supplied to the generation model.

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## 6. Metrics

Northstar monitors product and AI metrics separately.

OrbitFlow metrics include workflow success rate, execution duration, and connector failures. Beacon metrics include retrieval latency, generation latency, citation coverage, and unanswered-query rate.

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## 7. Engineering Practices

Changes normally use feature branches, automated tests, review, staging validation, and production deployment.

Unit tests cover business logic, integration tests cover service boundaries, and end-to-end tests cover critical journeys. AI changes are evaluated against versioned benchmark datasets.

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## 8. Customer Segments

Operations customers automate approvals, notifications, synchronization, and reports.

Technical customers use webhooks, APIs, service accounts, retries, rate-limit handling, and idempotency keys. User-delegated access should use OAuth 2.0.

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## 9. Support

Production support requests are categorized as incidents, service requests, or product questions.

Critical incidents receive an initial response target of 30 minutes during supported hours. Resolution time depends on impact and technical complexity.

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## 10. Glossary

A workflow is a saved sequence of triggers and actions. A run is one workflow execution. A connector integrates an external system.

RAG means retrieval-augmented generation. An embedding is a numeric representation of text used for similarity comparison.

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